# Challenges in the Everyday Lives of the Employed DHH Community in India

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One-sixth of the global DHH (Deaf or Hard-of-Hearing) population resides in India, constituting over 63 million people. Still, research on the DHH population has been mostly concentrated in the Global North. In this qualitative and quantitative study, we conducted 15 interviews and surveyed 131 people from the employed DHH community in India, to understand accessibility issues faced by them daily. We focus on the employed DHH community as they face challenges in a variety of settings (including home, workplace and transportation) and are usually the early adopters of assistive technology. Our work reveals key challenges faced by them, including non-availability of sign language interpreters at workplace, banks and hospitals, limitations of auto-generated captions for online meetings, reliance on hearing people to communicate thereby reducing their agency, and commute-related issues. We found workaround solutions, like hiring interpreter as supervisor, conversing over WhatsApp or by writing on a piece of paper, and using specific video conference platform. Some of the identified communication-related problems were associated with early speech-based education, role of parents, diversity in Indian sign language (ISL), and structural differences between ISL and English. Finally, we propose technology- and policy-related recommendations to improve accessibility for the employed DHH community in India.

CCS Concepts: • **Human-centered computing → Empirical studies in accessibility**; *Accessibility design and evaluation methods*; **User studies**.

Additional Key Words and Phrases: Accessibility, Individuals with Disabilities & Assistive Technologies, Workplaces, Empirical study that tells us about people

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## 1 INTRODUCTION

The Deaf or Hard-of-Hearing (DHH) community constitutes over 5% of the global population [4]. Due to the inaccessible nature of their environments, this community faces a variety of challenges in their everyday lives. Most of the accessibility research that studies these challenges are based in the Global North[1]. These studies explore the socioeconomic context of the lives of the DHH community, such as the role of family [37], communication challenges in educational institutions [62], and access to healthcare [53]. Prior work in resource-rich countries has also explored accessibility issues faced by DHH employees in a workplace setting—examining communication patterns, career barriers and difficulties faced by the DHH community in the Australian workplace [85], communication preferences of DHH professionals in supervisory roles in the USA [41], and a literature survey about experiences of stress and fatigue among the members of the DHH community at work in USA, Europe and Australia [83]. Other studies assume well-resourced settings and explore technological interventions to aid communication between DHH and hearing people, like custom camera

[1]Global North refers to the countries located primarily in the northern hemisphere that have historically been identified as "the West" or "first world" due to perceptions of their relative wealth, technology, and global dominance.

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setups to recognize sign language [3, 17], use of video/telecommunications relay services to make calls [8, 95], and teletypewriter technology on smartphones for emergency calls [99].

Many of these solutions and insights do not generalize to a country like India due to the large differences in cultural and economic context, access to technology, and employment opportunities. One example of this cultural difference is the nature of Indian Sign Language (ISL). Unlike American or British Sign Language, ISL is not a *single* language. It has a variety of dialects, such as Delhi Sign Language, Kolkata Sign Language, Bangalore-Chennai Sign Language, and Bombay Sign Language [1]. This diversity has impeded standardization and makes communication amongst the DHH community hard. The structure of ISL is also exclusive—the ISL lexicon has a high level of iconicity and compounding not seen in other sign languages [36].

Other factors specific to India include accessibility-related cultural taboos (reported in work on the Indian blind population [54, 56]), limited opportunities for education in mainstream schools [54], and the lack of trained interpreters [58]. The Indian DHH population also warrants its own study due to its large size—India is home to 63 million DHH people or one-sixth of the global DHH population [31, 42]. Additionally, hearing loss is the second most common disability in India, representing 18.9% of the disabled population [76]. Such a large community requires several studies in order to understand the accessibility challenges unique to them. Despite the above, a formal study on the challenges faced and potential support systems for the DHH community in India has not received much research attention.

In this work, we seek to address and focus on a part of this gap by asking the research question: *"What are the key communication challenges faced by the working Indian DHH community today?"* We focus on communication as it is a fundamental challenge faced by the DHH community [41, 85]. Additionally, we focus on the employed DHH demography, due to: (a) the importance of employment for earning a livelihood, developing skills, and collaborating with others [63], (b) challenges faced by them in a variety of settings including home, workplace, and commute, and (c) the fact that they are early adopters of assistive technology as they are financially independent [].

The DHH community has historically experienced higher rates of unemployment and lower levels of education compared to people without hearing loss [84]. Additionally, adults with hearing loss have a much higher unemployment rate in low- and middle-income countries compared to high-income countries [5], underlining the importance of such a study. With the recent push for diversity, equity and inclusion in the workplace [23], we believe technological solutions designed to make workplace accessible for the DHH community has higher likelihood of adoption.

In order to achieve our goal, we conduct both qualitative and quantitative studies of the employed Indian DHH community. Our qualitative study consists of 15 virtual interviews, facilitated by a certified sign language interpreter. Our quantitative study consists of 131 responses to an online form circulated amongst employed members of the DHH community. We find that the DHH community relies on workarounds such as writing or using basic signs to navigate through a variety of critical conversations. These workarounds have their limitations, forcing the community to rely on hearing people to help them carry out important conversations and thereby reducing their agency. Assistive technological solutions have either not been built or have not yet been adopted at scale by the community.

This work makes several contributions. First, we discuss the problems arising due to the stigma associated with sign language education, the grammatical inconsistencies in the English used by the DHH community, and the diversity of ISL. Next, we examine in detail the accessibility issues faced by the DHH community at the workplace and other everyday situations. Lastly, we contrast these challenges with those experienced by the employed American DHH population, and propose several inclusive technology design and policy recommendations that can improve accessibility in India.

## 2 RELATED WORK

In this section, we examine existing work in two related areas—global work on DHH-related accessibility and accessibility work in the Indian context. The latter has primarily focused on the visually impaired community, along with a few initial studies on ISL.

### 2.1 Research on the DHH Community across the Globe

The DHH community relies on assistive technology to understand speech. Simple aids such as using paper and pen to write back and forth or to text back and forth can work but are about 3-4 times slower than spoken or signed communication. Thus, they are not effective for sustained conversation [44]. Therefore there exists several technological solutions such as video/telephone relay services [8, 9], Internet Captioned Telephone Services [7], and Assistive Listening Devices [2].

The methods used by the DHH community to communicate with hearing people have been well studied [41, 47, 84] and uncover several communication-related workarounds including avoidance, lip reading, optimizing volume of speech, confirming the message, attracting the attention of the hearing person, and using simpler words and signs.

Communication challenges have also been studied in specific contexts. Significant work has been done in the context of education given its importance in shaping people's lives, such as the experiences of DHH students in educational institutions [62, 77], hearing students' perspectives on the inclusion of DHH students [28, 51] and methods to teach and assess DHH students [26, 27, 70, 73]. These studies have found that DHH students may not always disclose their needs and often settle for sub-par accommodations [25, 62]. They also dismantle misconceptions that technology can remove all access barriers or that technology built for one individual can generalize to others [62].

Another context that has been studied extensively is the home. The home plays a sensitive role in shaping the experience of DHH children, since more than 90% of DHH children are born to hearing parents [6]. Hearing parents of DHH children have to make several decisions regarding communication and language choices on behalf of their children. Research on the factors that influence this decision-making [34, 52] reveal that parents learn to view hearing loss in one of two ways: the *sociocultural* view of deafness aligning with the use of sign language, or the *audiological* view aligning with the use of the family's spoken language. The efficacy of these methods depends on the characteristics of each child. However, it has been observed that parents often choose a method based on inadequate information obtained from professionals and the priorities that they have for their children [98].

Lastly, an analysis of the workplace context is crucial due to its potential to support accessibility interventions. Several studies chronicle the current experiences of the DHH community in the workplace and propose and evaluate approaches to reduce the challenges faced by the community [38, 41, 71, 74, 82, 83, 85]. Quantitative findings from a survey-based study indicate that the most difficult workplace situations were meetings, training activities and work-related social functions [85]. There exist multiple accommodations to address these challenges, such as better lighting and furniture rearrangement to facilitate seeing other people better and access to sign language interpreters [71]. Apart from workplace provided accommodations, it is apparent that certain personal attributes like persistence and self-advocacy skills, as well as the presence of supportive and helpful co-workers are important to facilitate job satisfaction and success among the DHH community [35]. There has also been research on the awareness of the Americans with Disabilities Act (ADA)[2] [94] among employers, and the perspective of employers regarding the employment of DHH people [49].

[2]ADA is a comprehensive civil rights law in USA prohibiting discrimination based on disability in employment, state and local government programs, public accommodations, commercial facilities, transportation and telecommunications.

In recent years, there has been extensive research on the usability of Automatic Speech Recognition (ASR) with respect to the DHH community. For example, how ASR applications enhance communication exchanges between DHH and hearing people [44, 75] and how to best present ASR generated captions by using the system's word-level confidence, varying captioning styles, etc [15, 16, 61]. The studies showed that members of the DHH community use real-time speech-to-text, such as ASR, if its accuracy is least 85% [97] to 90% [66] or higher and the main factors that affect usability of ASR applications are noise (e.g., music), speech produced with an accent, multi-talker speech or side conversation and disfluent speech or speakers with emotion [44]. Another application is that ASR software is generally trained with speech samples from hearing people, which results in very poor recognition of deaf speech. This is because the acoustic and linguistic characteristics of speech associated with members of the DHH community is different from hearing people, and usually varies dramatically as a function of hearing loss and onset [78, 86]. As a result, devices that use speech-controlled interfaces are not usable by members of the DHH community [43, 45]. There has been some research on how to enable them to provide input to these devices by methods such as (i) human computation workflows for understanding deaf speech, and (ii) mobile interfaces that can be instructed to speak on the user's behalf. [21, 40]

Apart from communication related research on the DHH community, there also exists research about making content more accessible for the DHH community. An example of such research is automatic text simplification, which replaces unfamiliar words and complex grammar in literature with simpler sentences. [13, 57, 68]

### 2.2 Accessibility Research in India

Most of the above cited work on the DHH community has been localized to the Global North, where both the resources and awareness to address these challenges are relatively higher. In the wider space of accessibility research, significant work has been done over the past decade on the visually impaired community in India. This body of research has studied various aspects of visually impaired people's lives such as navigation [59, 60], education [55, 56], technology adoption and has proposed solutions [64, 87, 96]. Beyond the visually impaired community, there has also been research on developmental disabilities such as autism [12, 18, 89], cerebral palsy [91] and speech and motor impairment [33]. There is limited work on the DHH community in India. Existing work has primarily focused on educating DHH children [76, 100], methodologies to teach English to DHH students [39, 81], and teachers' perspectives on the education of DHH children [80]. We did not find any existing research that studies the broader set of challenges faced by the DHH community in India or that specifically focuses on the workplace context.

With respect to the Indian DHH community, the main focus has been on studying Indian Sign Language (ISL) [92, 93]. These studies focus on the diversity of ISL and its unique characteristics that differ from other sign languages and the many spoken languages of India. They examine the several regional dialects of ISL used across India [1], and its high iconicity and use of compound signs [36]. Another crucial feature of ISL that has been examined is its variation with class differences. Members of the educated middle-class DHH community use a unified and relatively standardized sign language, while the rural DHH have no exposure to this urban form. Instead, the rural DHH community use organically evolved *home-sign systems* [88], specialized to the socio-linguistic context of their communities. The Indian Sign Language Research and Training Centre[3] (ISLRTC) was established in 2015 to further research on ISL and promote its standardisation and usage. As part of a new National Education Policy (NEP) announced in July 2021, the government launched an ISL dictionary of 10,000 words and has made a decision to introduce ISL in the school curriculum to improve accessibility and create awareness [4].

[3]http://www.islrtc.nic.in/
[4]https://www.news9live.com/education-career/isl-dictionary-for-hearing-impaired-students-is-inclusive-120346

In summary, the challenges of the DHH community have mainly been studied in the Global North. However, many of these insights and technological solutions do not directly generalize to a country like India due to several reasons: (a) lower access to resources and education, (b) lack of accommodations for DHH employees and (c) distinctive features of ISL.

## 3 METHOD

We circulated an online survey amongst the employed DHH community in India, followed by semi-structured video interviews. Both the survey and the interviews had a focus on the workplace context but also contained general questions about accessibility issues in their everyday lives. The study was approved by the Institutional Review Board at <Anonymous Institute>. In this section, we present our survey and interview methodologies, along with our data collection and analyses techniques.

### 3.1 Survey

Our survey consisted of 18 questions—12 multiple-choice questions and 6 open-ended questions—and was administered via Google Forms.

The survey was organised in three sections. The first section consisted of multiple choice questions to collect demographic information (such as "*What is the level of your hearing disability?*" and "*Have you had speech therapy?*") and open-ended questions enquiring about the participants' gender, age, job, etc. The second section consisted of multiple choice questions based on communication challenges and workarounds (such as "*Where do you find it difficult to communicate with hearing people?*" and "*Which apps do you use to talk to your family and friends?*"). The final section consisted of multiple choice questions focused on the workplace context (such as "*Does your workplace have a sign language interpreter?*", "*How often do you have meetings at work?*", "*Do you use sign language while working?*", and "*Which apps do you use to communicate with your coworkers?*"). The final, optional question asked survey participants to provide their email ID and phone number in case they wanted to participate in an interview. The survey was administered in English. All the survey questions were short, simple and unambiguous to make them accessible to participants with low English proficiency. The surveys were piloted with two researchers and a certified ISL interpreter who provided feedback on question framing. The survey was accompanied with an explanation of the purpose of the survey in English as well as Indian Sign Language.

The survey form was distributed by a certified ISL interpreter, an NGO for the DHH community, and a philanthropic organization within their networks using WhatsApp and email. Furthermore, we asked the survey participants to share the survey within their WhatsApp groups. In order to be included in the survey, participants needed to be Deaf or Hard-of-Hearing (DHH), currently living in India, and be employed currently or in the recent past. The final optional question of the survey allowed participants to volunteer for the interviews, in which case they provided their email ID and phone number. No Personally Identifiable Information (PII) was collected unless the participants volunteered to participate in the interviews.

In total, we received 131 valid responses (107 male, 24 female, age=27.9±5.9 years) over 30 days spanning Jan–Feb 2021. All the survey respondents self-attested as DHH, out of which 42.2% of respondents have profound hearing loss (*i.e.*, complete deafness), 25.9% have moderate hearing loss and 31.9% have mild hearing loss. 24.4% of participants have graduate degrees, 49.6% have undergraduate degrees and the rest have a high school diploma.

The gender distribution of our respondents is 81.7% male, 17.6% female and 0.7% unknown. The mean age of respondents is $27.9 \pm 5.74$ years. 47.4% of our respondents have undergone speech therapy. 65.8% of respondents worked

from home, 22.2% of them worked solely from the office and another 12.0% worked in a hybrid model at the time of answering the survey.

### 3.2 Interview

We conducted semi-structured interviews after the survey during Feb–Aug 2021. Interview participants came from two predominant sources - participants recruited through our survey form , and friends and colleagues chosen by our sign language interpreter. Care was taken to ensure gender and occupation diversity. In total, we interviewed 15 participants (8 male, 7 female, age=21–34 years). 12 participants were recruited by our sign language interpreter and 3 participants were recruited from the survey. All participants were either currently or previously employed and lived in urban areas. Summarized information about the participants is available in Table 1.

In the interviews, we asked participants about their sign language education and proficiency, preferred mode of communication with hearing people (family, friends, coworkers and strangers), and workplace accessibility challenges and workarounds. Next, we asked them to recall recent difficulties faced by them in a public place or workplace and the approach they used to handle the situation. At the end of the interview, participants were invited to share open comments and express any concerns.

All interviews were conducted remotely by the first two authors using the Google Meet video conferencing tool, in the presence of a certified ISL interpreter (female, 27 years old). All the calls were video-recorded with the consent of the participants. Participants were informed that the data would only be used for research purposes and that no PII would be shared. The interviews lasted 30–75 minutes. The authors and the ISL interpreter interacted in English, while the ISL interpreter and the participant communicated in Indian Sign Language. The interviews were transcribed soon after they were conducted, and we use the exact translation provided by the interpreter when quoting participants. All participants' names have been anonymized in this paper. Participants were paid 500 INR for their participation.

We analyzed the interview transcripts using an inductive and iterative approach. All authors periodically discussed the interview data to identify emerging themes. We used an inductive, interpretive coding approach on this data [79]. Initial themes included 'communication with family', 'challenges concerning phone calls', 'lip-reading', and 'challenges in hospital'. We iterated over the data to produce higher-level themes including 'ISL education', 'work related challenges', and 'communication workarounds', and abstracted them out to the five main themes presented in this paper.

Please note that we refer to survey respondents as 'respondents' and interview participants as 'participants' for the rest of the paper.

## 4 FINDINGS

In this section, we discuss findings from our survey and interviews. We first describe the various accessibility challenges faced by the employed DHH community in the workplace. We then move on to describe accessibility challenges faced in other everyday scenarios. Lastly, we provide insights from our study that might explain the core reasons behind the above challenges - touching upon factors such as the state of sign language education in India, the role ISL plays in the way the employed Indian DHH population perceives English and the various forms of ISL. In each of the below subsections, we provide survey results along with our interview responses wherever applicable.

### 4.1 Work

We asked survey respondents the question - "*Where do you find it difficult to communicate with hearing people? (Select all that apply)*" with possible options being Work, Phone Calls, Bank, Family, Doctor, Transportation (while using Buses,

Table 1. Demographic details of the interview participants

| ID | Age | Sex | Hearing Loss Level | Location | Highest Education Degree | Sector | Occupation |
|---|---|---|---|---|---|---|---|
| P1 | 26 | M | Mild | Hyderabad | Under-graduate | IT/BPO | Movie editor |
| P2 | 26 | F | Moderate | Coimbatore | Under-graduate | IT/BPO | Security Guard |
| P3 | 37 | M | Moderate | Bombay | High School | Education | Teacher |
| P4 | 34 | M | Moderate | Hyderabad | Under-graduate | IT/BPO | Expenditure Auditor |
| P5 | 25 | M | Profound | Theni | Post-graduate | Retail | Data Entry Operator |
| P6 | 25 | M | Profound | Coimbatore | Under-graduate | E-commerce | Warehouse Assistant |
| P7 | 29 | M | Profound | Coimbatore | Post-graduate | IT/BPO | Process Executive |
| P8 | 21 | F | Profound | Hyderabad | Under-graduate | NGO | Video Creator |
| P9 | 25 | F | Profound | Hyderabad | Post-graduate | IT/BPO | Customer Support |
| P10 | 24 | F | Profound | Hyderabad | Under-graduate | IT/BPO | Expenditure Auditor |
| P11 | 32 | F | Profound | Delhi | Under-graduate | Education | Teacher |
| P12 | 33 | F | Profound | Hyderabad | Under-graduate | IT/BPO | Expenditure Auditor |
| P13 | 26 | M | Moderate | Trivandrum | Under-graduate | Education | Teacher |
| P14 | 25 | M | Moderate | Trivandrum | Under-graduate | IT/BPO | Software Developer |
| P15 | 21 | F | Profound | Hyderabad | Under-graduate | NA | Currently Unemployed |

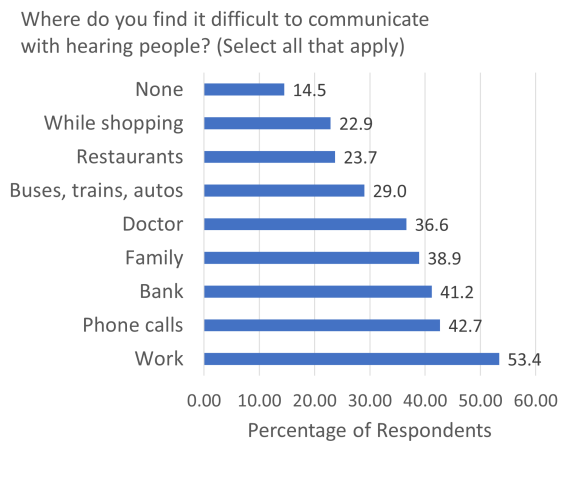


Fig. 1. Survey results for scenarios where DHH find it difficult to communicate

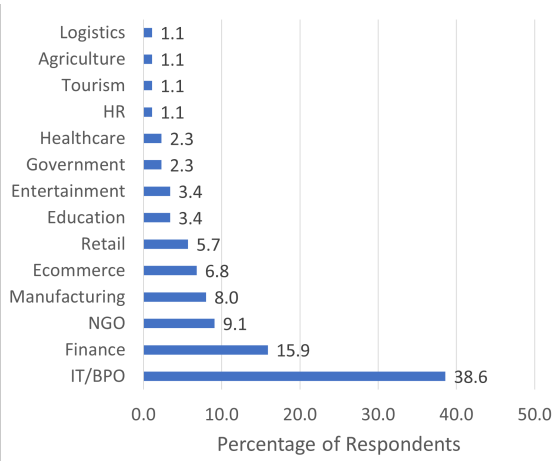


Fig. 2. Sectors that survey respondents worked in

Trains or Autos), Restaurants, Shopping or None (Figure 1). The results show that Work (53.4% of respondents) is the largest challenge area, followed by Phone Calls (42.7%), Bank (41.2%), Family (38.9%), Doctor (36.6%), Transportation (29.0%), Restaurants (23.7%) and lastly, Shopping (22.9%). The None option was chosen the least number of times (14.5%). Thus, workplace related challenges emerged as the key communication issue faced by the employed DHH community.

*4.1.1 Job Descriptions and Preparation.* Our survey was filled by participants working at 53 different companies. To gain a deeper understanding of sectors that employ the Deaf community, we mapped these 53 companies to 14 sectors (Figure 2). We found that the IT-BPO (Information Technology-Business Process Outsourcing) and Finance sectors

are the most common sectors that employed our survey respondents (The IT-BPO sector employed 38.6% and Finance employed 15.9% of respondents respectively). We note that this data may not be an accurate representation of DHH employment across India as the survey was distributed using snowball sampling with the initial participants from the IT-BPO sector. Our interview participants worked in 6 of these sectors, with a majority of them (8) employed by the IT-BPO sector. Respondents of the IT-BPO sector mainly worked jobs such as software engineers (13.7%), expense auditors (13.0%), and data entry operators (6.1%). Interview participant P4 described his job as an expense auditor in an IT-BPO company:

> "There is an app called Concur. We approve bills there. Bills come from third party, and we see if the amount being claimed matches the amount on the bill. If the amount is not matching, we have to manually delete it and enter the amount printed on the bill. Then we get to the next bill. We have to check each bill within 10 seconds, otherwise our reports per hour goes down. We have to reach [at least] 97% of the weekly target... They measure the bottom performer of each week. We have to login for 8 hours per day, with 1 hour of break." – *P4*

We found that such job descriptions were typical as they were designed to minimise communication with other colleagues or customers and worked around English literacy issues. 99.2% of survey respondents worked in internal contributor roles - roles wherein an individual can contribute to the growth of the organisation with minimal coordination with other colleagues. All our interview participants worked Individual Contributors roles as well. P6 stated that he was *"unable to grow"* in leadership roles in his organization due to communication challenges. The one exception to this was P8, who did not face communication challenges as she worked for an NGO consisting mainly of DHH employees

To further understand the type of work done by our survey respondents, we categorised each job into 2 classes: white collar and blue collar jobs. White collar jobs focus on knowledge work with opportunities for hierarchical advancement [72]. E.g., software engineers, teachers, and accountants. Blue collar jobs are physically labor intensive with stationary hierarchical level (typically in low-ranked positions) [72]. E.g., data entry operators, warehouse workers at e-commerce companies, and restaurant staff. We found that 61.4% of our survey respondents work in blue collar jobs and 38.6% in white collar jobs. The IT-BPO and finance sectors are the major employers of white collar jobs, while manufacturing and e-commerce sectors mainly employed blue collar workers.

Lastly, despite already having a college degree, 10 of our interview participants completed additional training courses to get employed. These courses are conducted by NGOs who teach skills like writing in English, typing on keyboard, using Microsoft Excel, and video editing. Our participants found these courses to be valuable—it taught them the in-demand skills and connected them with prospective employers. Thus, these additional courses significantly increase their likelihood of getting employed.

*4.1.2 Access to Interpreters at Workplace.* 49.6% of survey respondents stated that they have an interpreter present at work, while another 8.5% had interpreters present only during major events. The rest of them (41.9%) did not have interpreters at their workplaces. However, 76.3% of survey respondents said that they used sign language while working, indicating that sign language is preferred even if interpreters are not present at the workplace. Six of our interview participants stated that their office lacked an interpreter, but 4 of them still used ISL at their workplace. This was made possible using interesting workarounds. E.g.,

> "We don't have any interpreter. Initially we faced difficulty in talking to the supervisor but later we started teaching them sign language. We started with alphabets and a few words. Now they can sign! There is also another deaf girl who can talk, which makes things easier" – *P2*

*4.1.3 Meetings.* Meetings are a crucial component of office life, with employees across organisations attending meetings for an average of 6 hours/week [69]. 30.2% of our survey respondents reported having meetings everyday, with another 34.1% having meetings at least once a week. 19.0% reported never having meetings, while the rest (16.7%) would have them very rarely (1-2 times a month). For our interview participants, these meetings comprised of weekly performance discussions for each employee, rewards for good performance or training for future assignments.

When asked the question "*Are you able to talk to others during meetings?*", 53.7% of survey respondents responded with 'yes', with another 24.8% responding 'sometimes' and the rest (21.5%) stating 'no'. To understand these challenges better, we asked our interview participants about their experience in such meetings.

We found DHH teams with an interpreter are more successful at conducting meetings. In the organisations of P4, P10 and 12, the interpreter jointly leads the team with another manager. Thus, the interpreter has both work context and is comfortable with sign language. In case of P9's organization, the interpreter does not have a management role. She conducts a separate follow-up meeting with the DHH employees after a meeting concludes, to reiterate everything that was discussed.

Other participants (like P5) who did not have access to interpreters reported that they were unable to participate in meeting discussions. P5 would mostly skip meetings and rely on meeting minutes prepared by the team leader. While this workaround allowed DHH participants to be updated about the discussion, it precluded them from meaningfully contributing.

A few participants use lip reading to follow meeting discussions. For instance, P7 uses hearing aids to attend meetings. Having learnt lip reading and received speech therapy during his childhood, P7 can listen and respond to others as conversations happen during a meeting. However, this approach could get frustrating, as P7 described:

> "I feel bad when others are speaking fast and I am unable to tell them to slow down. If someone is talking to me fast and I am unable to understand then I ask someone else to explain what that person is saying. That person usually summarizes it instead of telling me the whole story and treats me like a baby... I don't like that. I sometimes turn off my video and sit because I don't understand what people are saying." – *P7*

Five participants mentioned trying the auto-generated captions feature offered by video calling applications (like Google Meet). While they found the feature useful, they stated 2 critical drawbacks that prevented them from using live captions regularly. First, participants struggled to read and understand captions when someone spoke fast; *"If a person is speaking too fast... only 50% of the time I can understand, rest of it I miss." – P1.* The other drawback was that the caption generator often make errors in understanding Indian accents and pronunciations, especially for Indian names.

*4.1.4 Conversations with managers.* Communication with one's manager is vital to success at work. In some offices with interpreters, the interpreter acted as the team lead for DHH employees. Although the employees had another reporting manager, their single point of contact was the interpreter, who resolved the DHH employees' doubts and raised concerns with the reporting manager as required. Communication between DHH employees and hearing managers is challenging in offices without interpreters. The default mode of communication is exchanging written messages (on a piece of paper, or over email/WhatsApp). While writing enables limited communication, it has two drawbacks. First, it is slow and becomes impractical for long, two-way conversations. Second, participants were concerned that writing about an issue

"*appears excessively formal*" and may be taken out of context as a complaint against the manager. E.g., P6 described his hesitation with writing an email requesting promotion:

> "Initially, my [DHH] seniors helped me understand and do my role. Now I want to go to the next level... I don't like doing the same work for a long time. But [my] manager is not able to understand what I'm trying to say. If I have to write and ask about the promotion, they [the management team] will be asking questions... I am afraid that they will give me lecture on it, so its better to not talk about it." – *P6*

Apart from written messages, two participants stated that they took the help of their hearing co-workers often to communicate with their manager. Our participants were more comfortable in having a written exchange with their co-workers, because the co-workers were not as busy as their managers and hence our participants could have a relaxed informal conversation with them. P14 (who is the sole Deaf software engineer in his organization) said the following when asked about whether he and his manager have ever discussed a salary hike:

> "When I need to have some conversation with my manager, I will first talk to a co-worker. Then he/she will come with me to explain to the manager. I used to write [on paper] and explain to co-worker for long conversations. Now I use WhatsApp to communicate [with the co-worker]... Conversations are usually about work-related issues. I never spoke about my salary with my manager." – *P14*

Other participants like P6 reported having limited conversations with his manager:

> "Before I joined [the team], there were Deaf people working already. They taught the manager sign language, but not much. I cannot talk about my job role to my manager. I can only say hi/bye to him. He [my manager] only understands conversations like 'did you have your tea?'." – *P6*

P6 works at an e-commerce warehouse that does not employ an interpreter. His job is to pack and seal products into boxes before they get shipped to customers. Due to his blue-collar work profile, he finds it difficult to communicate over WhatsApp/email.

*4.1.5 Conversations with co-workers.* All participants reported that the frequency and quality of interactions with their hearing colleagues were poor. E.g.,

> "I have only hearing colleagues at work. Conversations with my hearing friends won't go beyond simple greetings like 'Hi', 'Bye', etc. Sometimes the CEO meets me, and when he does he will ask 'How is your work?', 'How are you doing?' etc. During these times I would have communication problems and so I taught a few signs to the CEO." – *P14*

Such interactions happen using simple, intuitive signs that hearing co-workers can understand without any sign language training/knowledge.

In offices with interpreters, the interpreters make meetings accessible. However, they are usually not present for informal, unscheduled water cooler conversations. This results in minimal social interaction for the DHH employees.

Results to the the survey question *How do you communicate with your colleagues at work?*), show that WhatsApp (81.7%) is the most common app used to communicate with co-workers, followed by email (66.4%) and various video conferencing tools - Zoom (48.1%), Microsoft Teams (25.2%) and Google Meet (23.7%). While Zoom is the most commonly used video conferencing tool, this may only reflect their organization's adoption of specific tools and not necessarily the accessibility of these tools. Coincidentally however, our interview participants preferred Zoom (even in offices using Teams/Meet), as Zoom allows users to quickly shift between participant video tiles on a call (even on a mobile device), thus enabling DHH users to focus on the person speaking or signing. Google Meet on the other hand, automatically

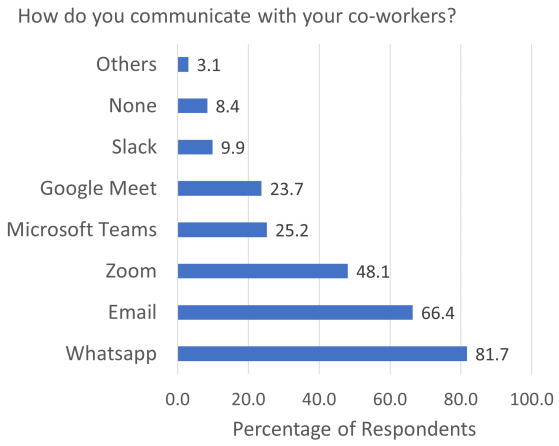


Fig. 3. Responses to the question "How do you communicate with your co-workers at work?". The responses have been normalised by the total number of participants taking part in the survey

identifies the speaker and prominently displays the speaker's video on every user's screen by default, which our participants found "*unusable as an interpreter was also on the call*". This was also reported as a challenge when viewing shared content such as slide decks which also occupy majority of the screen real-estate by default. Finally, this accessibility challenge also manifests in recordings of meetings where the content recorded depends on automatic choices made by the tools which are not necessarily informed by accessibility for the DHH community.

### 4.2 Other Everyday Scenarios

While workplace challenges are the most pressing, our survey results show that the DHH community face communication challenges in a variety of everyday scenarios. Here, we detail insights from the scenarios mentioned in Figure 1, going in descending order of scenario difficulty.

*4.2.1 Phone Calls.* Our participants never make audio calls. However, they occasionally receive audio calls. These calls are generally from food delivery personnel (e.g., Swiggy, Zomato), e-commerce delivery personnel (e.g., Amazon, Flipkart) or ride-hailing drivers (e.g., Uber, Ola). The calls are of 2 types—confirmatory calls informing the service requester that they have arrived at their doorstep and query calls asking for guidance to reach the exact address of the service requester. Addresses in India are not well-defined, hence such query calls are not uncommon [20].

> "When I book a cab, it is difficult to guide the person to my place. If I go to the doctor for fever or any problem, I can at least write and show them. But with drivers, there is no way to make them understand. Sometimes I text, but they don't reply and keep calling." – *P9*

The query calls cause distress to our participants as they can neither understand nor reply to the callers. Five participants reported taking their hearing neighbours' or co-workers' help to answer such calls on their behalf. The COVID-19 pandemic has resulted in a steep rise in the usage of home delivery services, thus exacerbating this accessibility problem.

*4.2.2 Banks and Post Offices.* Participants complained that bank and post office officials did not understand sign language, but hiring an interpreter to accompany them while visiting these places is prohibitively expensive. Their only

option was to take help from a hearing person. However participants like P8 were hesitant to "*disturb*" their family members and friends for every task. The combination of lack of interpreters, crowded bank, and complex procedures makes it a challenging scenario:

> "I went to the bank to change my ATM card. There was already a queue at the counter, so I did not have the time to write and show to them and explain... It was difficult... I went to the bank in Delhi. They say you have to go to Kerala [a state in South India] and do this work. There was a lot of communication problem with hearing people. They asked me to get someone from my family to assist me. It was my work, but I was not able to communicate with him [the bank official]. Also many people were there, so he can't spend a lot of time with me. I was trying to explain to them that I had filled a form for a new ATM card because I wanted to change it. They were not able to understand me and asked me to come back with my mother." – *P13*

Even commonly used workarounds like writing on a piece of paper do not work, due to the time needed for a two-way written conversation. Participant P3 reported bank managers "*getting angry*" at him for the delays caused in writing-based communication, as he was "*holding up the queue*". Despite these difficulties, P10 reported that her DHH primary school taught her the kind of questions to ask at banks and similar institutions. With this training, she is able to manage independently.

*4.2.3 Home.* We found that our interview participants faced challenges in communicating with members of their family. In particular, 7 participants stated that their parents do not know sign language, and that conversations with their parents are limited to simple questions and gestures, like "*Did you eat your food?*". A few participants (like P1) are able to lipread their parents. Parents often discourage the use of sign language due to the stigma associated with it and encourage their children to speak. Such social stigma against people with disabilities has been known to result in significant challenges [65].

> "I was in a boarding school from 1st to 10th [grade], where we had a sign language interpreter... My parents talk a lot. I taught them few signs, but they don't like me signing, so they always speak. If I don't understand then they use sign language. I don't have much conversation with them. My parents tell me to practice speaking even if I don't understand it... I don't like to interact with them much." – *P12*

Three participants were able to communicate with their parents in sign language—two of them have Deaf parents and the third taught sign language to his hearing parents.

Our participants reported that their siblings were more receptive to learning sign language. Four participants have hearing siblings who know "*ISL really well*" and with whom they can have complete conversations. For conversing with their parents, most participants took the help of their siblings (or a friend) who can both sign and speak.

In response to the survey question *"What app do you use to talk to friends and family?"*, data shows our survey respondents used WhatsApp most commonly (95.4% of respondents), followed by Zoom (58.0%), Google Meet (30.5%) and Microsoft Teams (23.7%).

*4.2.4 Hospitals.* Our participants found visiting a doctor to be accessible, as they are generally "*patient and educated*", and agree to have a written conversation. However, miscommunication with a doctor can be potentially risky as it can lead to misdiagnosis and/or incorrect medication. Hence, the DHH community prefers to take the help of a hearing family member, friend, or an interpreter for doctor visits, particularly for serious illnesses.

> "Hospitals are good, no problem. The nurse is also there, they help us, they can read. So if it is a small thing like fever then I can write and show. But if I have a major thing, then I have to bring an interpreter for better understanding. I don't know how to simplify the language and explain. It's better to take someone who knows sign language, as it's risky and a big issue if we don't understand what they [the doctor] are saying. For example, what if we go and try to explain to doctor but we still have the doubt of whether we should take the medicine or not. So it is better to take an interpreter or your family member." – *P13*

Moreover, as per P10, the sentence structure and English grammar used by the DHH community increases the chances of miscommunication in a written conversation.

*4.2.5 Transportation.* We found that participants use 3 modes of transportation for their daily commute—cabs, auto-rickshaws, and public buses. As discussed in Section 4.2.1, ride-hailing cab drivers call their passengers for confirmation and address-related queries. All our participants have experienced their rides being cancelled by the driver when they did not pick up the driver's phone calls. Attempts by our participants to interact with them over text messages were mostly unsuccessful, as drivers were usually driving when our participants made the phone call and hence were unable to check and reply to messages. This inability to communicate with cab drivers emerged as a strong pain-point with 9 participants complaining about it. The most common solution was to take the help of a nearby hearing person to interact with the driver.

P4 stated a workaround, wherein his organization provides a free pick-up and drop service at a fixed time for its DHH employees. If the employee does not show up on time for his/her pickup, the driver calls the office interpreter and the interpreter checks with the DHH employee.

Despite the phone call related challenges, our participants preferred app-based ride-hailing services over hailing auto-rickshaws on the road: "*it is still easier to book a cab or an auto on the app than to catch one on the road as it is difficult to explain the destination to auto drivers. In ride-hailing apps, it is easy to put the [destination] address in text.*" – P1. Moreover, a large fraction of auto-rickshaw drivers do not understand English, which makes written message-based communication infeasible.

Lastly, our participants face unique challenges while using public buses. Public buses are the most affordable means of transportation in India and hence tend to be overcrowded. Moreover, the rider needs to know the correct bus stop to get down at, and the name of the next stop is usually announced by the bus conductor. Thus, a bus ride requires communication with the bus conductor or co-passengers, which is difficult using written messages, both due to the crowd and co-passengers often being unable to read English. For instance:

> "While travelling in the bus when I ask 'Which place is this?', they cannot understand and sometimes ignore me. When I write and ask them, people don't understand English because they are from village. That's why I have to write in Tamil and show them the paper." – *P7*

*4.2.6 Shops.* We inquired about 2 types of in-person shopping—frequent purchases such as buying groceries and occasional purchases such as shopping for clothes. Our participants preferred visiting large supermarkets over small grocery stores, as these supermarkets permit self-service, with prices listed on all products and aisles clearly labelled with their product categories.

In contrast, for clothes shopping, our participants preferred local smaller cloth markets due to the cheaper prices. Cloth markets have multiple shops selling nearly identical items. Participants were comfortable shopping there as their

interactions were limited to asking for the price. Moreover, the presence of multiple shops meant that if a storekeeper was unable to communicate well with them, they could move to the next shop. For instance:

> "I never used to go shopping as my mother always bought clothes for me... After coming to Hyderabad [where I live alone], I started going shopping with [DHH] friends. Now, I buy my own dresses. If I do this sign *[does the sign for money]*, the shopkeeper will show their fingers *[holds up fingers to indicate the price of the product]*. If the price is high, I bargain. I will try to write down my price and negotiate. If they don't understand, I will just leave and go the next shop. I don't face any problems." – *P15*

### 4.3 Insights into Indian Sign Language

In this subsection, we present findings about Indian Sign Language that we learnt over the course of our study. Specifically, we examine three things - the state of sign language education in India, the influence of ISL on the way the DHH community perceives English and the difficulties in communication due to the diversity of ISL.

*4.3.1 Sign Language Education.* A majority of our participants (11) attended schools for DHH children. The other 4 participants attended mainstream schools and struggled with social stigma throughout.

The participants who attended mainstream schools said that they did so as their parents did not want them to be seen as "different". They recalled that they were given a hearing aid at school and asked to sit in the front, in an attempt to ensure that they could hear and lipread the teachers.

> "I studied with other hearing students near my house. I went to NISH for college where I learnt sign language. It was difficult for me while studying in school. I sat in the first row in class. I used hearing aids and could understand. Sometimes I could not understand some words and kept asking the teacher. If it gets cold like during the rainy season, then my hearing reduces and I could not hear anything at all." – *P14*

This approach resulted in negative experiences—P2 and P12 did not understand their teachers at all and relied solely on notes written by their friends, while P14 stated that he faced social isolation in school as a result of his disability.

Among the participants who attended schools for DHH children, P1 mentioned that he was explicitly forbidden from using sign language at school. He recalled that he was not allowed to converse with teachers in sign language and was encouraged to lipread and speak instead. Similarly, P10 mentioned:

> "I can read lips a little, but I mostly use ISL. I don't speak... When I was a child, the teacher used to teach me and force me to speak. I went to a deaf school where there was a speech therapist, but I could not understand him." – *P10*

A key finding from our survey results is that this encouragement to speak is widely prevalent in India - 49.6% of our survey respondents responded 'yes' to the question *"Have you had speech therapy?"*. P4 shared that he was encouraged to "*practice speaking slowly and clearly*" by his parents, as they feared that he would be socially isolated if he relied only on sign language. Despite this, there was an overwhelming consensus amongst our interview participants that they preferred sign language to speech and that they were more comfortable with using ISL.

Five of our participants did not learn ISL at home or school but picked it up informally from peers, by watching videos on YouTube, or through apps like the DEF-ISL app[5]. Popular YouTube channels (gained from our survey where we asked the free-text question - "*What sign language channel do you watch on YouTube/Facebook?*") include ISH News, MBM Vadodra, NAD, The Daily Moth, Yunikee, ISLRTC, NIOS and others. Interview participants stated that they found

[5]https://play.google.com/store/apps/details?id=in.eightfolds.deafenabled

peer learning to be the the most effective way to learn ISL. Three participants mentioned clearing doubts by asking their friends and said that practising with friends gave them the confidence needed to use sign language in public. These participants later learnt ISL formally in college as a part of their undergraduate studies or by completing a sign language diploma course.

*4.3.2 English literacy.* As mentioned earlier, the use of ISL when communicating with the hearing community is often not feasible due to the lack of interpreters. Moreover, these interpreters are very expensive (costing 13.7–20.5 USD/hour). Thus, written English becomes the dominant method used while communicating with the hearing community. However, our participants face various communication challenges while writing and reading English. Below are a few examples of the unique sentence structure used by the Indian DHH community, provided by our interpreter from her WhatsApp conversations with DHH friends:

> "I go start my exercise walking now it" instead of "I shall now start my walking exercise".
> "He is Venkatesan exercise time short so lazy" instead of "He is Venkatesan, who's exercise time is short and hence he is so lazy."
> "Before I wash shirt and jeans" instead of "I washed my shirt and jeans before this".

Finally, P13 (a teacher) noted that DHH students often had difficulty in understanding idioms such as "*time flies*".

*4.3.3 Diversity of ISL.* As covered previously in Section 2.2, ISL is not a single, standardised sign language. This variation in ISL became a frequent topic of conversation during our interviews as our interpreter (who is from Hyderabad) would sometimes fail to understand signs used by our participants who were from different parts of India. An example of this variation is the follows:

> "The sign for 'marriage' is shown by a mangalsutra (an auspicious necklace used in Hindu weddings) *[performs a sign that draws out a necklace]* in Chennai, while in Hyderabad it is shown by the holding of hands *[touches right hand fingers with left]* .. just like when the bride and groom hold hands during Hindu weddings. Similarly in North [India] the sign for mother is *[touches her nose]* which indicates the nose ring while in Hyderabad you need to do two signs for mother - the North Indian sign followed by this *[holds fist to cheek]*." – *Interpreter*

We observed that these variations in sign language are reflective of India's cultural and linguistic diversity. India is comprised of 29 culturally-diverse states and is home to 184 languages (each of which is spoken by more than 10,000 speakers) [46]. ISL, just like spoken languages, is strongly influenced by the culture of its signers. Three of our participants mentioned knowing multiple signs for a given word, as they had grown up in multiple cities. P9 stated that even signs for basic words like '*mother*' and the days of the week varied across the different regions she grew up in.

Apart from the inconvenience, participants reported instances where these variations in ISL caused problems during lectures, training workshops, and workplace meetings.

> "Sometimes she [my manager at work who knows sign language] will stop and say 'I can't understand what you are signing.' She'll ask me twice or thrice again. She won't understand as she is from North [India]. Sometimes I get fed up and just text her to explain." – *P1.*

Apart from the diversity of ISL itself, 4 participants also reported the mixing of American Sign Language (ASL) with ISL. This influence of ASL on ISL was observed in participants from tier-1 cities like Delhi and Bombay. This phenomenon is similar to code-switching, wherein a speaker alternates between 2 or more languages in the same

conversation or utterance [14]. In particular, we found the ASL-ISL switch similar to the Hindi-English switch (called *Hinglish*), which is commonly observed among Hindi-speaking Indians [67]. Our participants specified that one key reason for code-switching is that ASL has a unique advantage. Unlike the two-handed ISL, all ASL alphabets can be signed using one hand [30], thus enabling them to sign while holding objects with the other hand. All 4 participants mentioned using ASL mainly for WhatsApp video calls as they could hold their phone with one hand and use the other hand for signing. While these participants used ASL for such scenarios, all of them reported that they were not fluent in ASL and could only fingerspell the alphabets and a few other basic signs. Other participants who did not use ASL would sign by using their body or face as a substitute for one hand while holding objects.

## 5 DISCUSSION

Our survey and interviews highlight key challenges faced by the Indian DHH community on a day-to-day basis. In this section, we examine key findings in the broader context of existing work and provide design and policy recommendations.

### 5.1 Agency

In an effort to gain agency and avoid stigma, our participants were encouraged by their teachers and parents from early childhood to speak, lip-read, use hearing aids, take speech therapy, and avoid signing in public. These methods were prevalent in schools for DHH children. While such approaches worked with moderate success for two hard of hearing participants, all profoundly deaf participants were unable to and/or uncomfortable with using these methods. Thus, well-meaning efforts by members of the hearing community to increase the agency of DHH individuals by reducing stigma actually have the opposite effect. The role of stigma in influencing decision-making amongst the guardians of children with disabilities has been previously reported, particularly in the Indian Blind population [54]. They found that parents did not provide white canes to their Blind children to avoid stigma and make their children '*look less blind*' [54], which curtailed their children's physical and mental growth. This suggests that social stigma supersedes agency and independence in India and serves as an additional hurdle for people with disabilities.

We found that the agency of a DHH individual varies across scenarios. A few factors influencing agency are the nature of interaction, the time available for communication and the English comprehension ability of the hearing person. For activities like shopping and ordering food at restaurants, we found our participants independently interacting with the hearing person using a combination of basic signs and written messages as the required interaction was short and there was time available. These interactions may fail if the hearing individual lacks the ability to comprehend English. Our participants faced difficulties in independently communicating with a driver or a doctor, as drivers usually have no time to have a written conversation and conversations with a doctor have a higher risk of misinterpretation. In such scenarios, the DHH person took the help of an interpreter or another hearing person (friend, family or co-worker). However, this reduces their agency and raises privacy concerns. Privacy may not be a major concern for short conversations with strangers (such as with a driver or a shopkeeper), but can be more relevant when conversing with a manager at work or while interacting with a doctor about a stigmatized health condition.

### 5.2 Workplace Accessibility for DHH employees: India versus the US

Foster et al. [41] detail challenges faced by the American DHH population at the workplace, focusing on Deaf supervisors. In contrast, none of our 15 interview participants and only 1 (out of 131) survey respondents were in a supervisory/managerial role. This suggests that more senior roles are not common for the DHH community in India. This

may be be attributed to the lack of resources in the Global South, for instance in providing access to full-time interpreters. The role of the interpreter is in sharp contrast across the Global North and South—in the US, most interpreters were secretaries of Deaf employees [41], while our participants reported that most interpreters were managers or colleagues. This suggests a power equation which may result in limited agency of the DHH employees in India. With respect to technology, as early as 2003, DHH employees in the US had access to relay services and TDDs (Telecommunications Device for the Deaf) [41]. TDD is an electronic device for text communication over a telephone line. However, none of our participants reported having access to comparable service/technology even in 2021. In our interviews, we found our participants relying minimally on technology, primarily limited to WhatsApp video calls and messages. When communicating with co-workers, we found similarities between Indian and US DHH employees with respect to the usage of written notes and emails for crucial messages, and minimal informal conversation or chit-chat. Lip-reading was found to be commonplace in the US, while in India, only one participant knew lip-reading perhaps due to the large number of spoken languages in India. With respect to signing, a few hearing co-workers in the US had even learnt ASL, which is in contrast to India where none of the participants reported colleagues who formally learnt ISL. This may be again due to the difference in job seniority of the DHH employees in US and India. Please refer to Table 2 for a summary of these similarities and differences.

Table 2. A comparison of our key findings at the workplace with a similar study conducted in the United States.

| Result | Findings from Foster et al. [41] | Our Findings |
|---|---|---|
| Job Role | In supervisory roles at work. | In internal contributor roles. |
| Interpreters | Easy access to interpreters. | Minimal access to interpreters. Limited to meetings. |
| Interpreter Position | Interpreters are a secretary to the DHH employee in most cases. | Interpreters are a manager of the DHH employee in most cases. |
| Phone Calls | Access to TDD and relay services. | No support to make phone calls accessible. |
| Lip Reading | Used lip-reading to converse with hearing people. | Only one participant could lip-read. |
| Writing | Emails and written notes. | WhatsApp, emails and written notes. |
| Chit-chat | Left out of informal conversations at work. | Left out of informal conversations at work. |
| Signing | A few co-workers learnt sign language to communicate with their DHH manager. | Signing was limited to simple signs which did not require formal sign language knowledge. |

### 5.3 Designing Inclusive Technology

**Interpreters on Demand**: Access to an interpreter emerged as the most reliable solution for the Deaf community. All of our participants mentioned the ease of interacting with hearing individuals in the presence of an interpreter, and mentioned a reduction in anxiety during such conversations. However, accessing interpreters at all times is infeasible due to the high cost of hiring an interpreter, the dearth of certified interpreters in India, and the inability of interpreters to be physically present on demand. Technologies like Video Relay Services solve these problems. Using a relay service has multiple advantages–it is cheaper, enables an interpreter to serve more people (as they can switch calls), and allows DHH users affordable on-demand hiring. While this technology is well developed in the Global North, it is not widely available in India. The closest to a video relay service we found is an app called SignAble[6] which was recently launched in 2020 in India. However, none of our participants were aware of it. We identify a strong requirement for a video relay

[6]https://play.google.com/store/apps/details?id=org.signable.apprtc

service in a variety of scenarios, and believe that such an app would be potentially useful if effectively advertised to the Indian DHH community.

**Video/Audio Calling Tools**: Video calling was used often by our participants in multiple scenarios, with an increase in usage due to COVID-19. The applications used for video calling varied—WhatsApp is used with friends and family, and Zoom, Google Meet, and Teams are used at work. In these applications, speech-to-text (STT) technology is used to auto-generate captions and serves as a partial technological alternative to an interpreter, which our participants found useful for work meetings. However, they experienced two issues while using the captions: (a) inaccuracy in generated captions as the STT tools often were inaccurate in understanding speech with Indian accents, and (b) their inability to read the generated captions at the required pace due to their limited proficiency with English. A simple solution to address this would be to allow a history of the caption that the DHH people can revisit. A more technically challenging solution would be to integrate automatic tools for text simplification [90] such as Lexi [22]. Another challenge with video conferencing tools is the use of the screen real-estate. Tools like Google Meet default the primary video on each user's screen to the current speaker based on audio signals. This design choice makes the tool inaccessible to the DHH community, who would like to see the interpreter prominently at all times. This was also reported as a challenge when viewing shared content such as slide decks and in meeting recordings ( 4.1.5). To address these challenges, video conferencing apps can provide more agency to DHH users by allowing manual configurations for what is laid out on the screen and in recorded content. Apart from the video calling applications, a few apps on the Google Play Store (such as Rogervoice[7]) attempt to make audio phone calls accessible for the DHH community by extensively using STT and TTS (text-to-speech) technologies. Rogervoice auto-generate captions for incoming audio calls, and provides an optional pre-recorded message to inform the caller that they are speaking to a DHH person. The DHH person can either talk to the caller or type out a message that is read to the caller using TTS technology. Unfortunately, none of these apps are available in India. We recognize that the technological barriers to such an app are higher in India given the diversity in languages and accents, and presence of code-mixing.

**Ambient Conversation**: Our participants struggled to take part in informal social conversations with co-workers. Similar challenges were reported among DHH employees in the US [41]. Even listening to ongoing conversations while not contributing can help DHH employees connect with their co-workers. Certain apps exist that enable DHH users to listen to ambient conversations. For instance, Talk to Deaf[8] uses Google's STT technology to display captions for conversations happening around the user's smartphone. However, we found that the app worked only when a person is speaking directly into the microphone and when there is minimal ambient noise, severely limiting their utility. Improved microphone technology for indoor settings is available on many consumer-grade smart speaker devices which can add support to help the DHH hear ambient conversations.

**Accessible Service Apps**: While efforts to build applications that improve accessibility continue, several problems faced by the DHH community can perhaps be solved by designing existing applications more inclusively. In particular, a majority of our participants faced challenges in interacting with cab drivers and e-commerce/food delivery personnel over phone calls, and have to seek help from family, friends, neighbors, co-workers or strangers. These service apps may add a "Do not disturb mode" or "DHH mode" informing the driver/delivery person that the service requester is a DHH individual. In such cases, communication can be over messages instead of phone calls.

**"Broken" English**: The poor English sentence structure seen in 4.3.2 is probably due to a difference in underlying sentence structure between ISL and English. While English follows the SVO (subject - verb - object) order for arranging

[7]https://play.google.com/store/apps/details?id=com.rogervoice.app
[8]https://play.google.com/store/apps/details?id=unique2040.com.text2speech

words in a sentence, ISL follows the SOV (subject - object - verb) order [92]. Due to this, the DHH community tends to follow the SOV order even while communicating in English, resulting in sentences that are confusing to read. A similar phenomenon has been observed in other countries as well. For instance, Italian Sign Language has an underlying SOV structure whereas the Italian language follows the SVO order, which results in non-native signers of Italian Sign Language following the SVO order while signing [29]. The other major difference between ISL and English is the absence of articles (like 'a' or 'the') and connectives (like 'and' or 'or') in ISL [19]. This results in missing articles/connectives when DHH people write in English too. A tool that inputs text in ISL-like SOV structure and generates text in SVO structure with appropriate grammar checks may address this accessibility gap. Moreover, the output text can be translated to local languages using existing Neural Machine Translators (NMTs), potentially resolving challenges faced by the DHH community as reported while using public transportation (Section 4.2.5).

**Conversion between ISL and text**: Automatically converting sign language videos, as captured by a smartphone camera, to natural language text is a challenging task. However, this is an active area of research in the AI community and several recent papers have made progress [24, 32, 50, 93]. Like other AI tools, training a sign language recognition system requires large amounts of training data, which consists of sign videos and their corresponding text translations. Given its diversity, a large and concerted effort is required to create such datasets for ISL. Further, such data collection effort may initially be focused on the most sensitive contexts, such as communication at the workplace, with service providers, at banks, and while travelling. Finally, such datasets must be released in the open-source to enable AI researchers to build sign language recognition systems. Integrating such a tool would be most useful in communication apps like WhatApp, which is used by 95.7% of survey respondents for personal calls and 81.7% of them for professional calls. This would require that the sign language recognition tool be compatible with low-to-medium end smartphone cameras.

### 5.4 Policy Recommendations

While technological solutions can help address some of the issues mentioned in the Findings section, many others are structural in nature and require policy intervention. For example, the taboo associated with using sign language in schools (mentioned in Section 4.3.1) was exacerbated by the lack of official recognition for an Indian Sign Language subject. This was remedied in July 2021 by the National Education Policy [11] that gave ISL the status of a language subject. The policy also calls for the standardisation of ISL while allowing regional variants to be taught, thereby alleviating some of the problems that arise due to ISL's diversity (Section 4.3.3). Another laudable initiative on improving accessibility in schools is the Haryana government's translation of state school textbooks to Indian Sign Language [48]. The government's Central Board of Secondary Education should consider releasing similar translations for all textbooks.

According to our survey results (mentioned in Section 4.1.2), 41.9% of our respondents do not have access to interpreters at work. Section 4.2 also reveals the absence of interpreters from other important places such as hospitals and banks. Policy intervention and training is required to increase the number of interpreters and guarantee access to them for DHH individuals. Mandates to provide content in an accessible manner, such as the Ministry of Information & Broadcasting's memorandum to service providers to deliver subtitles/closed captioning/sign language options across news television programmes [10] are also a welcome measure. As noted in Section 4.3.2, steps should be taken to ensure that the language used in these captions is simple enough to be understood by the DHH community.

Another influential step towards ensuring access to interpreters and content in Indian Sign Language would be the recognition of ISL as an official language of India in the Indian constitution, which would encourage its development, preservation and enrichment.

## 6 LIMITATIONS

This study provides a rich description of the lives of the employed DHH community in India. There are a few limitations to this study which the reader should consider when determining how these findings may generalize to other contexts:

- While the introduction to the survey was circulated in Indian Sign Language, the survey itself could only be accessed in English. Thus, this limited the audience of the survey to employed DHH respondents who could read and write in English. Since the survey was targeted at the employed DHH community, it is reasonable to assume English literacy amongst our target respondents
- Due to the variation and diversity in Indian Sign Language, our interpreter often had difficulty understanding some participants during our interviews. The issue was mitigated as much as possible by finger-spelling and repeatedly clarifying what the participant was saying
- Some demographic info about survey participants is absent (such as their ISL variation preference)
- The study included relatively few interview participants, all of whom were employed. The interview participants were recruited using snowball sampling and therefore the representativeness of the sample is not guaranteed.

## 7 CONCLUSION

In this work, we study the challenges faced by the employed DHH community through interviews with 15 DHH participants and 131 DHH survey respondents. We emphasize the specificity of our study here as a reminder to readers that this study is at best a first step towards characterizing accessibility challenges for the DHH community outside of the developed regions context. Our study reveals the various challenges faced by the Indian DHH community on a day-to-day basis. First, we discuss foundational challenges due to the stigma associated with signing in India, the English literacy issues faced by the DHH community, and the problems arising due to the variations in ISL, which is characteristically different from other sign languages like ASL. Next, we cover context-specific challenges such as the lack of interpreters in the workplace, the inability of the DHH population to converse with managers and hearing co-workers, and accessibility challenges in other common places such as the home, banks and hospitals. Lastly, we cover technology related challenges, where we highlight problems with video calling applications and automated captioning services, and discuss the issues with app-based service delivery. We end by contrasting the Indian workplace challenges with the experiences of DHH employees in the Global North, and propose technology and design recommendations to tackle the three types of identified challenges.

## 8 ACKNOWLEDGEMENTS

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